Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

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Abstract

Between 70–96 % of maritime casualties involve human error [1], and adverse weather compounds it by degrading both perception and the very communications crews rely on to coordinate avoidance. We present a tick-based maritime agent-based model on the real Baltic and North Sea, in which vessels follow hand-authored shipping tracks and resolve encounters through a Closest-Point-of-Approach (CPA) detector coupled to a COLREGs-style give-way rule. The contribution is a quantified causal chain from *environmental and human degradation* to *collision outcome*: the success of every collision-warning exchange is gated by the product of local weather severity and a circadian crew-fatigue scalar, so tired watch-keepers in heavy weather miss warnings and collide. Collision consequences are evaluated with an accredited probabilistic damage model (DTU SIMCOL), external collision frequency is cross-validated against the regulator-grade IWRAP Mk II model, and a search-and-rescue subsystem layers IAMSAR random-search detection (MASSIM) with month-conditioned Arctic environmental degradation (Ashrafi). Against a classical and a weather-aware baseline, the system is evaluated across four scenarios with 30 seed-matched runs per condition using Mann–Whitney U tests with Holm–Bonferroni correction and Cliff's δ effect sizes. We hypothesise that fatigue-aware watch scheduling reduces the fatal-event rate by $\geq 20\,\%$ and improves post-incident survival by $\geq 15\,\%$ relative to the classical baseline.

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport carries roughly 80 % of global trade, yet the sector's safety record is dominated by a persistent human-error problem. The International Maritime Organization (IMO) attributes 70–96 % of all incidents to human causes [1], and the European Maritime Safety Agency (EMSA) identifies weather as a compounding factor in a disproportionate share of total-loss events, particularly in congested coastal corridors where crew fatigue and information latency interact [2]. Hull and machinery insurance claims exceed USD 3.5 billion annually [3]; even a 5 % reduction in fatal incidents would yield hundreds of millions in avoided liability.

The International Regulations for Preventing Collisions at Sea (COLREGs) [4] provide a deterministic collision-avoidance baseline whose implicit assumption — that crews perceive the encounter geometry accurately and communicate intentions reliably — fails under two compounding conditions that prior agent-based maritime simulation [5, 6, 7] rarely couples directly to outcome:

- **Environmental degradation of communication.** Reduced visibility, sea clutter, and heavy weather degrade the bridge-to-bridge (VHF) exchanges through which two converging vessels agree who gives way. Most collision-avoidance models assume that, once a risk is detected, the manoeuvre is always executed.
- **Human-factor coupling.** Crew fatigue accumulates over watch cycles and follows a circadian pattern with a late-night performance nadir [8]. Prior models isolate fatigue in peripheral sub-models without a direct, quantified connection to whether a warning is acted upon.

The contribution of this work is a maritime agent-based model in which the success probability of every collision-warning exchange is the *product* of an environmental factor and a crew-fatigue factor, creating a quantified causal link from *information quality under stress* to *collision and survival outcomes*. We build the model on the real Baltic and North Sea geography (Section 3), drive vessels along hand-authored shipping tracks, and resolve encounters with a CPA/TCPA detector and a COLREGs-style give-way rule (Section 5). Collision consequences use the accredited DTU SIMCOL probabilistic damage model (Section 6); the resulting collision frequency is cross-validated against IWRAP Mk II (Section 7); and a search-and-rescue subsystem (Section 8) closes the loop from incident to survival. The engine is implemented in Rust on the krABMaga agent-based framework [9] for deterministic, data-parallel batch execution. Figure 1 summarises the causal pipeline that ties these components together.

2. Research Objectives

The central thesis is: *coupling crew-fatigue dynamics to the reliability of collision-warning communication, and managing fatigue through smart watch scheduling, reduces fatal maritime outcomes more effectively than weather-unaware classical practice.*

Experimental conditions. All experiments compare three methods, implemented as configuration switches over a single agent codebase:

Baseline A (Classical) represents current weather-unaware practice: vessels follow COLREGs give-way with idealised (perfect) VHF communication, no weather-induced slowdown, and an unmanaged fatigue build-up.

Baseline B (Shore Rest Alerts) adds weather-aware speed reduction and shore-issued rest advisories that slow fatigue accumulation, but no further mitigation.

Proposed System adds smart watch scheduling (lowest fatigue build-rate) on top of the weather-aware mitigations of Baseline B. Performance is measured using six

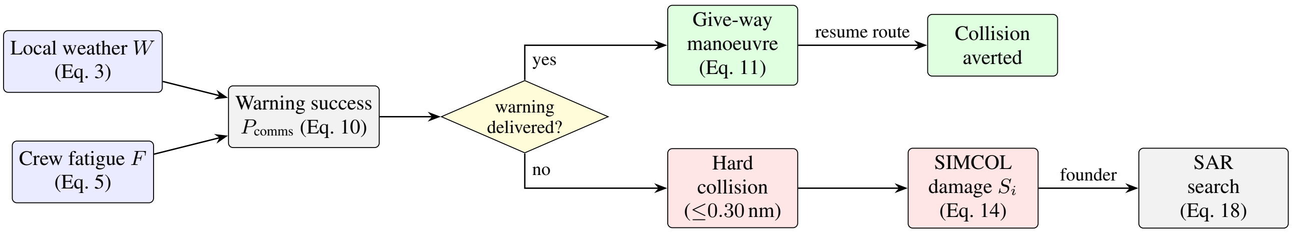

Figure 1: *Causal pipeline of the model. Local weather W and crew fatigue F jointly set the collision-warning success probability P_{comms} (Eq. 10); a delivered warning yields a give-way manoeuvre, a lost warning leaves the encounter to geometry and may produce a hard collision. Collision consequences are evaluated by the SIMCOL damage model, and a foundering vessel enters the search-and-rescue chain.*

KPIs defined in Table 2.

Falsifiable hypotheses. They drive all design decisions; any component that contributes to none of them is excluded.

H1. Fatigue-aware scheduling (Proposed) reduces the fatal-event rate (Table 2, KPI 1) by $\geq 20\,\%$ versus Baseline A ($p < 0.05$, Mann–Whitney U, Cliff's $|\delta| \geq 0.2$).

H2. Earlier, better-coordinated avoidance and a functioning rescue chain improve post-incident survival (Table 2, KPI 4) by $\geq 15\,\%$ versus Baseline A.

H3. The Proposed System outperforms Baseline B on the fatal-event rate ($p < 0.05$) under the storm-corridor scenario, where comms degradation is most severe.

3. System Model

3.1. World Geometry and Projection

The simulation occupies the real Baltic and North Sea basin, bounded by latitude $\phi \in [50.5^\circ, 66.0^\circ]$ N and longitude $\lambda \in [-5.0^\circ, 31.0^\circ]$ E — from the English Channel and Dutch coast in the south-west to the Gulf of Finland in the east and the mid-Norwegian coast in the north. The domain is mapped with an equirectangular projection carrying a cosine-of-mid-latitude correction, so that one field unit equals exactly one nautical mile on both axes at the reference parallel $\phi_0 = \frac{1}{2}(\phi_{\min} + \phi_{\max}) = 58.25^\circ$. With $k_{\text{lat}} = 60\,\text{nm}/^\circ$ and $k_{\text{lon}} = 60\cos\phi_0$, a geodetic point projects to field coordinates

$$x = (\lambda - \lambda_{\min})\,k_{\text{lon}}, \qquad y = (\phi - \phi_{\min})\,k_{\text{lat}}. \qquad (1)$$

The domain measures 930 nm north–south ($15.5^\circ \times 60$) and $\approx$1136 nm east–west; the residual east–west distortion is at most $\sim$6–10 % at the box edges and is accepted without correction terms, so the engine treats all distances as Euclidean nautical miles. The coastline is loaded from Natural-Earth polygons into an R-tree spatial index supporting accelerated point-in-polygon (land) and segment-intersection (grounding) tests.

3.2. Time Resolution

Time advances in discrete ticks of $\Delta t = 15$ min (0.25 h). The default horizon is $T = 2880$ ticks (30 days). A wall-clock-of-day variable, initialised at 06:00 UTC, advances 0.25 h per tick and drives the circadian fatigue term (Section 4). All stochastic draws use a single per-run seeded generator (`SmallRng`), so identical seeds reproduce byte-identical KPI logs.

3.3. Agent Types and Voyage Logic

The model contains vessel agents and, on incident, rescue agents (Section 8). Each vessel i carries position $\mathbf{p}_i$, an ordered route $R_i = (\mathbf{w}_0, \ldots, \mathbf{w}_{m-1})$ of waypoints, a route index and travel direction $d_i \in \{+1, -1\}$, heading ψ_i, cruise speed v_i, a discrete state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew count $n_i = 10$, and an internal fatigue scalar $F_i \in [0, 1]$.

Routes are *hand-authored* ordered key-point polylines drawn in a browser-based editor (`tools/waypoint_editor.html`) and stored as `data/ais_paths.json` with schema `mmsi/name/vessel_type/waypoints[]`. Fixing the lane geometry by hand — rather than replaying raw AIS — gives a controlled, reproducible traffic substrate for the information-quality experiment while preserving realistic basin-scale routing. Ports are likewise hand-authored polygons (`data/ports.json`); a vessel that enters a port polygon docks.

Cruise speed is drawn at spawn by vessel type ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases} \qquad (2)$$

Waypoint pursuit. With active target $\mathbf{t}$ (an avoidance waypoint if queued, else the current route waypoint), $\boldsymbol{\delta} = \mathbf{t} - \mathbf{p}_i$ and $\rho = \|\boldsymbol{\delta}\|$, a waypoint is reached when $\rho < 0.5$ nm. The per-tick step is $s = \min(v_i\,\Delta t,\ \rho)$ and the proposed move is $\mathbf{p}_i' = \mathbf{p}_i + (\boldsymbol{\delta}/\rho)\,s$, with compass heading $\psi_i = (90^\circ - \operatorname{atan2}(\delta_y, \delta_x)) \bmod 360^\circ$. To prevent tunnelling through thin landmasses, the move is accepted at the first of up to five geometrically halved step lengths $s \cdot 2^{-j}$ ($j = 0, \ldots, 4$) whose segment does not cross land; a post-move grounding guard reverts $\mathbf{p}_i$ to the last valid position if it ended on land. On reaching a port or route endpoint a vessel docks for a randomised dwell $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses ($d_i \leftarrow -d_i$). A fixed fleet size N is maintained: when $|\text{fleet}| < N$, a fresh vessel is spawned onto a random hand-authored track.

3.4. Weather and Hazard Field

A 50×50-cell grid evolves three physical channels — sea state s, visibility v, and wind u, each in $[0, 1]$ — which are fused into a scalar hazard $W_{ij} \in [0, 1]$:

$$W = \operatorname{clip}\left(\frac{w_s\,s + w_v\,(1 - v) + w_u\,u}{w_s + w_v + w_u},\ 0,\ 1\right), \qquad (3)$$

with weights $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. A vessel samples W at its current cell only (nearest-cell lookup), so each vessel acts on local conditions. The field advances each tick through seven ordered stages (full parameters in Section 10.4): (i) a two-state calm/storm regime Markov chain selecting mean-reversion targets; (ii) up to eight Lagrangian storm cells with growth, decay, drift and removal; (iii) an AR(1)-with-advection background per channel; (iv) a system overlay imprinting storm cells onto the channels; (v) cross-channel coupling; (vi) 5-point smoothing with a gradient limiter; and (vii) hazard fusion via Eq. 3.

Independently of the stochastic field, the **storm corridor** scenario carries a single moving storm zone of radius r_{storm} that drifts at 0.30 nm/tick along a fixed lat/lon track (English Channel → North Sea → Skagerrak → Kattegat → Baltic → Gdańsk). Inside the zone, both communication reliability (Section 5) and vessel speed (Section 10.3) are degraded.

4. Crew Fatigue Dynamics

Each vessel carries an internal fatigue scalar $F_i \in [0, 1]$ (0 rested, 1 exhausted), updated every tick. The circadian drive peaks near 03:00 and troughs near 15:00:

$$c(t_{\text{utc}}) = \tfrac{1}{2}\left(1 - \cos\left(\tfrac{2\pi(t_{\text{utc}}-3)}{24}\right)\right) \in [0, 1]. \tag{4}$$

While ACTIVE, with local hazard W, fatigue accumulates as

$$\Delta F = \left(\beta_{\text{base}} + \beta_W\, W + \beta_c\, c(t_{\text{utc}})\right) m(\text{method}), \tag{5}$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015,\ 0.0040,\ 0.0008)$ and a method build-rate

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A (unmanaged)} \\ 0.82 & \text{Baseline B (shore rest alerts)} \\ 0.65 & \text{Proposed (smart scheduling)}, \end{cases} \tag{6}$$

and $F_i \leftarrow \min(1,\ F_i + \Delta F)$. While DOCKED, the crew recovers at $F_i \leftarrow \max(0,\ F_i - 0.025)$ per tick (full recovery in ≈40 ticks). The method factor m is the sole lever distinguishing the three conditions' human-factor behaviour, isolating the effect of fatigue management.

Fatigue → communication degradation. Tired watch-keepers miss or delay warnings. With threshold $F^* = 0.40$ and maximum penalty $\kappa = 0.65$,

$$g(F_i) = \begin{cases} 1, & F_i \le F^* \\ 1 - \kappa\,\dfrac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35,\ 1]. \tag{7}$$

This factor multiplies into the collision-warning success probability (Eq. 10), the mechanism through which fatigue translates into collisions.

5. Collision Avoidance

5.1. CPA / TCPA Detection

Each ACTIVE pair is screened with the standard linear-motion closest-point-of-approach result. The field-frame velocity for heading ψ and speed v is $\mathbf{v}(\psi, v) = v\,\Delta t\,(\sin\psi,\ \cos\psi)$. For a pair (a, b) with relative position $\Delta\mathbf{p} = \mathbf{p}_b - \mathbf{p}_a$ and relative velocity $\Delta\mathbf{v} = \mathbf{v}_b - \mathbf{v}_a$,

$$t^* = -\frac{\Delta\mathbf{p}\cdot\Delta\mathbf{v}}{\|\Delta\mathbf{v}\|^2}, \qquad d_{\text{CPA}} = \|\Delta\mathbf{p} + t^*\,\Delta\mathbf{v}\|, \tag{8}$$

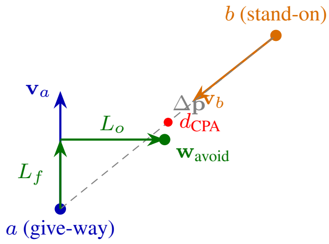

Figure 2: *Give-way geometry. The lower-id vessel a turns to starboard toward a temporary waypoint $\mathbf{w}_{avoid}$ placed L_f ahead and L_o to starboard of its heading (Eq. 11), opening the closest-point-of-approach distance d_{CPA} to the stand-on vessel b. The manoeuvre executes only if the warning is delivered (Eq. 10).*

with t^* measured in ticks. Degenerate cases return $(d_{\text{CPA}}, t^*) = (\|\Delta\mathbf{p}\|, \infty)$ when $\|\Delta\mathbf{v}\|^2 < 10^{-9}$ (parallel) and $(\|\Delta\mathbf{p}\|, 0)$ when $t^* \le 0$ (diverging). A manoeuvre is considered iff *all* of

$$\|\Delta\mathbf{p}\| \le 20\text{ nm}, \quad d_{\text{CPA}} \le 5\text{ nm}, \quad 0 < t^* \le 12\text{ ticks}, \tag{9}$$

hold and the give-way vessel is not in avoidance cooldown.

5.2. Give-Way Rule and Communication Success

The vessel with the lower identifier is the give-way (turn-to-starboard) vessel; the other stands on. A deterministic lower-id rule is a reproducible surrogate for the COLREGs crossing/head-on stand-on/give-way assignment [4, 5]. Crucially, the manoeuvre is executed *only if the warning is successfully communicated*, with success probability the product of an environmental factor and a fatigue factor over both vessels:

$$P_{\text{comms}} = \underbrace{\min\left(q(\mathbf{p}_a),\ q(\mathbf{p}_b)\right)}_{\text{weather / storm}} \cdot \underbrace{\min\left(g(F_a),\ g(F_b)\right)}_{\text{fatigue, Eq. 7}}, \tag{10}$$

where $q(\mathbf{p})$ is the nominal link reliability, reduced to the storm-comms rate (as low as 0.08 inside the storm corridor — 92 % of warnings lost) when $\mathbf{p}$ lies in the storm zone. The manoeuvre proceeds iff a uniform draw is below P_{comms}; otherwise the give-way vessel fails to act, leaving the encounter to resolve by geometry alone. Equation 10 is the model's central coupling: it makes warning delivery jointly contingent on weather and human fatigue.

5.3. Avoidance Geometry and Message Exchange

A give-way vessel receives one temporary waypoint placed $L_f = 3$ nm ahead and $L_o = 5$ nm to starboard of its heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f\,(\sin\psi,\ \cos\psi) + L_o\,(\cos\psi,\ -\sin\psi). \tag{11}$$

A three-message exchange is logged — `CollisionWarning{cpa,tta}` → `GivingWay` → `MaintainingCourse`, with `ResumeRoute` once the avoidance waypoint is consumed. Each issued warning records its t^* for the average-TTA KPI, and a 3-tick cooldown prevents manoeuvre thrashing.

5.4. Optional Continuous-Avoidance Track (Force Field)

As an optional realism layer, the discrete give-way waypoint of Eq. 11 can be replaced by a continuous artificial-force-field manoeuvre in the style of Xiao et al. [10], in which COLREGs behaviour emerges rather than being scripted. A vessel responds to a repulsive force from each nearby obstacle (vessel or shoal) within an effect distance d_o,

$$\mathbf{F}_{\text{rep}} = \sum_{k:\, d_k < d_o} \frac{k_{\text{rep}}}{d_k^2}\, \hat{\mathbf{n}}_k, \tag{12}$$

where d_k is the distance to obstacle k, $\hat{\mathbf{n}}_k$ the unit vector away from it, and k_{rep} a steepness constant. The resultant force sets a rudder angle that drives a first-order Nomoto manoeuvring response with non-dimensional indices $K' = K(L/v)$, $T' = T(v/L)$. The effect distance is calibrated from AIS encounter statistics, with $d_o \sim \mathcal{N}(1548,\, 706^2)$ m for head-on and $\mathcal{N}(384,\, 358^2)$ m for overtaking encounters [10]. In this track, the crew-preparedness state scales k_{rep}, so a fatigued crew yields weaker, later avoidance.

6. Collision Consequence Model (SIMCOL)

A **hard collision** is logged whenever two ACTIVE vessels close within $d_{\text{hit}} = 2\, r_{\text{coll}} = 0.30$ nm. The consequence of a hard collision is evaluated with the DTU SIMCOL probabilistic damage and survivability model [11], which replaces an ad-hoc casualty rule with collision mechanics.

External dynamics. The kinetic energy available for structural deformation in a collision is the loss of kinetic energy of the rigid-body system between first contact and the common-velocity state. For a striking ship of mass M_a (including added mass) at speed V_a impacting a struck ship at relative collision angle θ, the absorbed energy E_{abs} follows the closed-form external-dynamics solution of Pedersen and Zhang as summarised in [11, 12]; in the right-angle limit it reduces to the classical form

$$E_{\text{abs}} = \tfrac{1}{2}\, \frac{M_a M_b}{M_a + M_b}\, V_a^2, \tag{13}$$

where M_b is the struck ship's mass plus added mass.

Internal mechanics. The absorbed energy is related to damage extent through Minorsky's absorbed-energy/resistance-volume correlation [13], $E_{\text{abs}} = a\, R_T + b$, where R_T is the resistance (damaged) volume of deformed structural material. Lützen casts the resulting damage as *non-dimensional* distributions: the damage length ℓ/L_{pp} (longitudinal extent normalised by struck-ship length) and the penetration t/B (transverse extent normalised by breadth), each fitted as a probability density conditioned on the striking-ship displacement, speed, and collision angle [11].

Survivability. A sampled damage case (ℓ, t, location) defines a set of flooded compartments, from which the model assigns a survivability factor $S_i \in [0, 1]$ — the conditional probability that the struck vessel survives that flooding case (a residual-stability s-factor in the probabilistic damage-stability sense). The per-collision expected fatalities are then

$$\mathbb{E}[\text{fatalities}] = n_b\, \big(1 - S_i\big), \tag{14}$$

with n_b the struck-ship crew. A founder ($S_i \to 0$) triggers evacuation and the search-and-rescue chain (Section 8); a survived case ($S_i = 1$) is logged as a near-miss damage event.

7. External Collision-Frequency Validation (IWRAP)

The internal KPIs (Section 9) measure relative differences between methods but are not, by themselves, regulator-grade frequencies. To anchor the model to accepted practice, each run's vessel tracks are exported as synthetic AIS and analysed offline

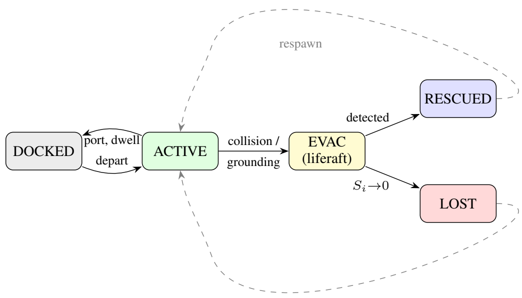

Figure 3: *Vessel state machine. A vessel cycles* ACTIVE ↔ DOCKED *along its route; a foundering collision or grounding sends it to* EVAC *(liferaft), from which the SAR chain yields* RESCUED *(search detection, Eq. 18) or* LOST *($S_i \to 0$). Terminal vessels respawn to maintain fleet size.*

with the IWRAP Mk II model [14]. The expected number of collisions on a waterway is the product of a geometric candidate count and a causation probability [15, 14]:

$$N_c = N_a\, P_c. \tag{15}$$

For two opposing flows of classes i, j over a segment of length L_W during time t, the head-on candidate count is

$$N_a^{\text{head}} = \frac{D\, t}{L_W} \sum_{i,j} Q_i^{(1)} Q_j^{(2)}\, \frac{(B_i^{(1)} + B_j^{(2)})\, (V_i^{(1)} + V_j^{(2)})}{V_i^{(1)}\, V_j^{(2)}}, \tag{16}$$

with traffic flows Q, beams B, speeds V; the overtaking form replaces $(V_i + V_j)$ with $|V_i - V_j|$. For crossing waterways meeting at angle $\Theta \in (10^\circ, 170^\circ)$,

$$N_a^{\text{cross}} = \sum_{i,j} Q_i^{(1)} Q_j^{(2)}\, \frac{D_{ij}\, V_{ij}}{V_i^{(1)}\, V_j^{(2)}}\, \frac{1}{\sin\Theta}, \tag{17}$$

where $V_{ij} = \sqrt{(V_i^{(1)})^2 + (V_j^{(2)})^2 - 2V_i^{(1)} V_j^{(2)} \cos\Theta}$ is the relative speed and D_{ij} the geometric collision diameter [14, 12]. The causation probability is taken from the established range $P_c \approx 0.5$–1.3×10^{-4}, with $P_c^{\text{head-on}} = 1.3 \times 10^{-4}$ and $P_c^{\text{crossing}} = 1.0 \times 10^{-4}$ [14, 15]. Comparing N_c between Baseline A and the Proposed System provides an externally interpretable, regulator-aligned check on whether the fatigue-coupling mechanism produces frequency reductions of the order expected for a real fairway (acceptance $\approx<1$ collision/yr/fairway).

8. Search-and-Rescue Subsystem

When a collision founders a vessel (Eq. 14, $S_i \to 0$) or a grounding forces an evacuation, the survivors enter a liferaft and the search-and-rescue (SAR) chain is triggered. A rescue agent is dispatched from the nearest shore base as either a helicopter (80 kn) or a patrol vessel (25 kn) [16], after a mobilisation delay $\Delta t_{\text{mob}} = 2$ ticks. The subsystem layers two accredited components onto the transit. Figure 3 shows the resulting vessel state machine.

8.1. Drift and Random-Search Detection (MASSIM)

Because the exact distress position is uncertain and the liferaft drifts, detection is modelled with the Koopman random-search formulation used in the MASSIM maritime-search model [17, 18]. A searcher of sweep width w moving at speed v_s over a search area A has instantaneous detection rate $\gamma(t) = w\,v_s/A$. The cumulative detection probability for n searchers over elapsed search time τ is

$$\mathrm{CDP} = 1 - \exp\Big(-\frac{n\,w\,v_s\,\tau}{A}\Big) = 1 - e^{-C}, \qquad (18)$$

where $C = n\,w\,v_s\,\tau/A$ is the search coverage factor. The search area expands as the datum drifts: with an initial uncertainty radius r_0 and drift speed u_d, the area grows as $A(\tau) = \pi\,(r_0 + u_d\,\tau)^2$, so detection becomes harder the longer the response is delayed — the quantitative link from response latency to survival. Liferaft drift advects the datum at the prevailing wind/current velocity each tick [17].

8.2. Environmental Degradation (Ashrafi)

Rescue-asset performance degrades with weather and season following the Arctic SAR framework of Ashrafi et al. [19]. Each cell/month draws weather parameters — wind speed, significant wave height, air temperature, and a wind-chill index — by Monte-Carlo sampling of historical data. The wind-chill index uses the standard form

$$\mathrm{wci} = 13.12 + 0.6215\,T - 11.37\,U^{0.16} + 0.3965\,T\,U^{0.16}, \qquad (19)$$

with air temperature T (°C) and wind speed U (km/h). Each parameter x is discretised into a severity level L_x by thresholds, and an aggregate severity $SL = f(L_{\text{wind}}, L_{\text{wave}}, L_{\text{wci}})$ is mapped — through an expert-judgement function $g_i(\cdot)$ — onto reductions in the rescue agent's travel speed and boarding speed, and onto hard no-go thresholds (e.g. icing or hover limits that suspend operations). The total rescue time is the sum over time-steps of transit, search (Eq. 18), boarding, and refuelling under the severity-degraded speeds. This replaces a linear weather penalty with month-conditioned severity bands and no-go states; in the source case study the longest rescue times occur December–February and the shortest May–August [19].

9. Experimental Design

9.1. Methods Under Test

The three conditions of Section 2 are realised as configuration switches over one agent codebase:

- **Baseline A (Classical):** weather-unaware; idealised VHF ($q \equiv 1$); no weather-induced slowdown; unmanaged fatigue ($m = 1.00$); collision-consequence multiplier highest.
- **Baseline B (Shore Rest Alerts):** weather-aware slowdown (Section 10.3); reduced fatigue build-rate ($m = 0.82$); weather/storm-gated comms.
- **Proposed System:** smart watch scheduling ($m = 0.65$) plus all Baseline B mitigations.

9.2. Scenarios

Table 1 summarises the four scenarios; each runs 30 independent seeds for 2880 ticks (30 days). With three methods and four scenarios, the full plan executes $4 \times 3 \times 30 = 360$ runs, dispatched in parallel.

Table 1: *Scenario summary. All scenarios use 30 seeds and 2880 ticks/run (30 days at 15 min/tick).*

Scenario	Key conditions	Ships	Hyp.
1 — Calm Passage	calm-biased weather; nominal comms. Calibration target <0.5 collisions / 1 k ship-hrs.	20	Calib.
2 — Storm Corridor	moving storm zone; comms rate as low as 0.08 inside.	25	H1, H3
3 — Blind Shore	reduced fleet; degraded coordination away from dense lanes.	25	H1
4 — Deep Water Rescue	sparse fleet; long rescue transits stress the SAR chain.	15	H2

9.3. Key Performance Indicators

Table 2 defines the six KPIs. Ship-hours accumulate 0.25 h per ACTIVE vessel per tick.

Crew preparedness (KPI 6) is accumulated each tick for every ACTIVE vessel as the product of a weather-readiness and a fatigue-readiness term, with method awareness factor $\alpha_M \in \{2.0\,(\mathrm{A}),\ 1.3\,(\mathrm{B}),\ 1.0\,(\mathrm{Proposed})\}$:

$$P_{\text{prep}} = \max(0,\, 1 - \alpha_M W) \cdot \max(0,\, 1 - 0.70\,F). \qquad (20)$$

9.4. Statistical Analysis

For every (KPI × scenario × method-pair), the two methods are compared with a two-tailed Mann–Whitney U test (average-rank tie handling, normal approximation): with $\mu_U = n_1 n_2/2$, $\sigma_U = \sqrt{n_1 n_2 (n_1 + n_2 + 1)/12}$, $z = (U - \mu_U)/\sigma_U$, and $p = 2\Phi(-|z|)$, where Φ is evaluated by the Abramowitz–Stegun 26.2.17 rational approximation. The effect size is Cliff's $\delta = (U_1 - U_2)/(n_1 n_2) \in [-1, 1]$ [20]. Multiplicity is controlled by Holm–Bonferroni step-down over all tests; a result is *significant* if $p_{\text{corr}} < 0.05$ and *confirmed* if additionally $|\delta| \geq 0.2$. Method pairs tested are (A vs Proposed), (B vs Proposed), and (A vs B). The same seed sequence is applied to all conditions, eliminating seed-selection bias from pairwise comparisons.

10. Implementation

10.1. Engine and Execution

The model is implemented in Rust on the krABMaga agent-based framework [9], with an axum/tokio REST + WebSocket server for live run control and a rayon-parallel batch runner; a React/Leaflet frontend renders vessels, ports, the storm zone, the weather grid, the comms log, and KPIs live. Batch runs write per-run JSONL tick logs, a manifest, and a `summary.csv` for analysis. Vessel agents are scheduled at

Table 2: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatalities ÷ ship-hrs $\times 10^3$	H1
2	`collision_per_1k_hrs`	Collisions ÷ ship-hrs $\times 10^3$	H1
3	`evac_activation_rate`	Evac activations ÷ ship-hrs $\times 10^3$	H1
4	`survival_ratio`	$1-$ fatalities / crew exposed in collisions	H2
5	`avg_tta_hours`	Mean rescue Time-to-Arrival across dispatches	H2
6	`mean_p_prep`	Time-averaged crew preparedness across vessels	H1, H3

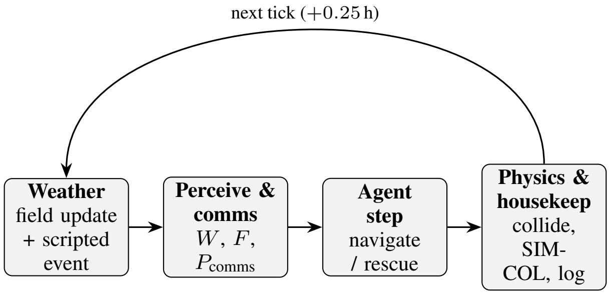

Figure 4: *The four ordered phases of each* 15*-minute macro tick. Data flows strictly left to right within a tick; weather precedes perception so a vessel acts on the field issued in the same tick, and physics/housekeeping closes the tick before the clock advances.*

krABMaga ordering 200 and a single post-tick agent at ordering 1000.

10.2. Simulation Loop

Each macro tick runs four ordered phases (Fig. 4) whose ordering is a causal constraint: (1) *weather* — advance the field and apply any scenario-scripted event; (2) *perception & communication* — each vessel samples local hazard, updates fatigue (Eqs. 4–7), and CPA-screened encounters attempt their warning exchange (Eq. 10); (3) *agent step* — vessels navigate at hazard-gated speed (Section 10.3) or hold for avoidance, and rescue agents decrement mobilisation or advance under the SAR penalty (Section 8); (4) *physics & housekeeping* — hard-collision and grounding checks, SIMCOL consequence evaluation (Section 6), terminal-vessel respawn, logging, and clock advance.

10.3. Speed Reduction under Hazard

Two multiplicative slow-downs are applied by interpolating the just-moved position back toward the last valid position by a factor f. The hazard-gated factor (Baseline B and Proposed only; Baseline A ignores weather) is

$$f_W = \max\big(0.30,\ 1 - 0.45\,W\big), \tag{21}$$

and a storm-zone factor $f_{\text{storm}} \in \{1.0\,(\text{A}),\ 0.45\,(\text{storm corridor}),\ 0.70\,(\text{default})\}$ damps speed inside the moving zone, acting as an implicit braking mechanism that lowers closing rates in deteriorating weather.

10.4. Weather Process Parameters

The seven-stage field of Section 3.4 uses: regime flip probabilities $p_{cs} = 0.03$, $p_{sc} = 0.08$ (Mixed preset; scaled $\times(0.35, 1.40)$ Calm, $\times(2.00, 0.50)$ Stormy); reversion targets Calm$(0.30, 0.80, 0.25)$ / Storm$(0.72, 0.30, 0.75)$; up to $K_{\max} = 8$ storm cells with intensity decay $\tau \in \{0.006\,\text{storm}, 0.020\,\text{calm}\}$, radius $r = 8(1 + 0.15\sin\theta)\sqrt{I}$, and removal at $I < 0.12$; an AR(1) background

$$X_{t+1} = \rho_X\,\text{adv}(X_t) + (1-\rho_X)\,\mu_X + \sigma_X(0.3+\mathfrak{u})\,\eta_t, \tag{22}$$

with $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$, unpredictability $\mathfrak{u} = 0.30$, and $\eta \sim \mathcal{U}(-1, 1)$; a system overlay $\omega = (1 - \sqrt{d^2/r^2})^2 I$ adding to s, u and subtracting 0.85ω from v; cross-channel coupling $s \mathrel{+}= 0.35(u-0.5) - 0.20(v-0.5)$, $u \mathrel{-}= 0.25(v-0.5)$; and a 5-point smoother with gradient cap $G_{\max} = 0.12$.

11. Discussion

The model's defensible core is the explicit, quantified coupling in Eq. 10: a single collision-warning exchange succeeds only when both the environment and both crews permit it. This makes the central hypotheses (H1–H3) mechanistic rather than correlational — fatigue management (m in Eq. 6) acts on outcomes solely by changing how often warnings are delivered and how prepared crews are, with no other lever distinguishing the conditions. Re-baselining the report on the real Baltic–North Sea engine, with hand-authored tracks and manual ports, keeps the experiment controlled while remaining geographically realistic.

The accredited layers raise the model from an internal A/B comparison toward external validity. SIMCOL (Section 6) grounds casualty counts in collision mechanics rather than a tunable rate; IWRAP (Section 7) places the resulting frequencies on a regulator-recognised scale; and the SAR subsystem (Section 8) makes `survival_ratio` and `avg_tta_hours` meaningful by modelling drift, random-search detection, and environmental degradation rather than instantaneous rescue. Several external formulas remain marked `TODO-VERIFY` where the source treatment is a distribution or an expert-elicited mapping rather than a closed-form constant; these should be pinned to the cited regulation or case study before final reporting.

Limitations follow from the controlled design. Hand-authored tracks trade empirical traffic realism for reproducibility; the deterministic lower-id give-way rule is a simplification of full COLREGs assignment; and the optional force-field track (Section 5.4) is not the default. Calibration targets — notably fewer than 0.5 collisions per 1 000 ship-hours under Baseline A in the Calm Passage scenario — are grounded in AIS-derived near-miss encounter statistics [6, 7].

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A. Default Constants Reference

Table 3 lists the principal engine constants with default values; all are overridable via `SimulationConfig` or the dashboard.

Table 3: *Selected engine default constants.*

Constant	Default	Role
Δt	0.25 h	tick length (15 min)
T	2880	ticks per run (30 d)
weather grid	50×50	hazard field
(w_s, w_v, w_u)	$(1.0, 0.5, 0.8)$	hazard fusion weights
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o	3 / 5 nm	give-way waypoint offsets
cooldown	3 tk	re-manoeuvre lockout
r_{coll}	0.15 nm	hit at $2r$ = 0.30 nm
$(\beta_{\text{base}}, \beta_W, \beta_c)$	0.0015, 0.0040, 0.0008	fatigue rates
recovery	0.025/tk	dock fatigue recovery
F^*, κ	0.40, 0.65	comms-degradation knee
storm comms rate	0.08	corridor link reliability
n crew	10	per vessel
dwell	8–48 tk	port stay
helicopter / patrol	80 / 25 kn	rescue asset speeds
Δt_{mob}	2 tk	rescue mobilisation delay
P_c (IWRAP)	0.5–1.3×10^{-4}	causation probability